

Springer Texts in Statistics

Robert W. Keener

# Theoretical Statistics

Topics for a Core Course



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# **Springer Texts in Statistics**

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Robert W. Keener

# Theoretical Statistics

Topics for a Core Course



Springer

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To Michael, Herman, Carl, and Billy



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## Preface

This book evolved from my notes for a three-semester sequence of core courses on theoretical statistics for doctoral students at the University of Michigan. When I first started teaching these courses, I used *Theory of Point Estimation* and *Testing Statistical Hypotheses* by Lehmann as texts, classic books that have certainly influenced my writings.

To appreciate this book students will need a background in advanced calculus, linear algebra, probability, and some analysis. Some of this material is reviewed in the appendices. And, although the content on statistics is reasonably self-contained, prior knowledge of theoretical and applied statistics will be essential for most readers.

In teaching core courses, my philosophy has been to try to expose students to as many of the central theoretical ideas and topics in the discipline as possible. Given the growth of statistics in recent years, such exposition can only be achieved in three semesters by sacrificing depth. Although basic material presented in early chapters of the book is covered carefully, many of the later chapters provide brief introductions to areas that could take a full semester to develop in detail.

The role of measure theory in advanced statistics courses deserves careful consideration. Although few students will need great expertise in probability and measure, all should graduate conversant enough with the basics to read and understand research papers in major statistics journals, at least in their areas of specialization. Many, if not most, of these papers will be written using the language of measure theory, if not all of its substance. As a practical matter, to prepare for thesis research many students will want to begin studying advanced methods as soon as possible, often before they have finished a course on measure and probability. In this book I follow an approach that makes such study possible. Chapter 1 introduces probability and measure theory, stating many of the results used most regularly in statistics. Although this material cannot replace an honest graduate course on probability, it gives most students the background and tools they need to read and understand most theoretical derivations in statistics. As we use this material in the rest



of the book, I avoid esoteric mathematical details unless they are central to a proper understanding of issues at hand. In addition to the intrinsic value of concepts from measure theory, there are several other advantages to this approach. First, results in the book can be stated precisely and at their proper level of generality, and most of the proofs presented are essentially rigorous.<sup>1</sup> In addition, the use of material from probability, measure theory, and analysis in a statistical context will help students appreciate its value and will motivate some to study and learn probability at a deeper level. Although this approach is a challenge for some students, and may make some statistical issues a bit harder to understand and appreciate, the advantages outweigh these concerns.

As a caveat I should mention that some sections and chapters, mainly later in the book, are more technical than most and may not be accessible without a sufficient background in mathematics. This seems unavoidable to me; the topics considered cannot be covered properly otherwise.

Conditioning arguments are used extensively in the book. To keep the derivations as intuitive and accessible as possible, the presentation is based on (regular) conditional distributions to avoid conditioning on  $\sigma$ -fields.<sup>2</sup> As long as the conditioning information can be viewed as a random vector, conditional distributions exist and this approach entails no loss of generality. Conditional distributions are introduced in Chapter 1, with the conditioning variable discrete, and the law of total probability or *smoothing* is demonstrated in this case. A more general treatment of conditioning is deferred to Chapter 6. But I mention in Chapter 1 that smoothing identities are completely general, and use these identities in Chapter 6 to motivate the technical definition of conditional distributions.

With advances in technology for sharing and collecting information, large data sets are now common. Large sample methods have increasing value in statistics and receive significant attention in this book. With large amounts of data, statisticians will often seek the flexibility of a semi- or nonparametric model, models in which some parameters are viewed as smooth functions. At a technical and practical level, there is considerable value in viewing functions as points in some space. This notion is developed in various ways in this text. The discussion of asymptotic normality for the maximum likelihood estimator is structured around a weak law of large numbers for random functions, an approach easily extended to cover estimating equations and robustness. Weak compactness arguments are used to study optimal testing. Finally, there is an introduction to Hilbert space theory, used to study a spline approach to nonparametric regression. Modern statisticians need some knowledge of functional analysis. To help students meet the challenge of learning this material, the presentation here builds intuition by noting similarities between infinite-

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<sup>1</sup> A reader with a good background in probability should have little trouble filling in any missing technical details.

<sup>2</sup> Filtrations and conditioning on  $\sigma$ -fields are mentioned in Chapter 20 on sequential analysis.

dimensional and finite-dimensional spaces, and provides motivation by linking the mathematical results to significant statistical applications.

If you are a professor using this book as a text, please note that results from Chapters 1 through 4 and Sections 6.1 and 6.2 are used extensively in the rest of the book, and any unfamiliar material on these pages should be covered with care. But much of the rest of the book can be resequenced or omitted to suit your preferences. Chapters 7 and 15 on Bayesian methods should be covered in order, as should Chapters 12, 13, and 17 on hypothesis testing. Chapter 11 on empirical Bayes estimation uses results from Chapter 7, and Chapter 14 on the general linear model uses results on testing from Chapter 12. Results on large sample theory from Chapters 8 and (to a lesser extent) 9 are used in Chapters 15 through 20. As I mentioned earlier, results in some chapters and sections<sup>3</sup> are more mathematically challenging; depending on the maturity of your students you may want to omit or cover this material superficially, possibly without proofs or derivations. For these chapters and sections and others, title footnotes indicate whether the material is optional and how the results will be used later.

Finally, a few words of appreciation are in order. To Michael Woodroffe, Herman Chernoff, and Carl Bender, who have had such an impact on my personal development as a mathematician, probabilist, and statistician; to friends, family, colleagues, and the Department for support and encouragement; and to past students, reviewers, and editors for a wealth of useful suggestions. This manuscript was typeset using L<sup>A</sup>T<sub>E</sub>X and figures were produced using MATLAB. Finally, a special thanks to future students; the notion that this book will help some of you has kept me believing it to be a worthwhile project.

Ann Arbor, Michigan  
June 24, 2010

ROBERT KEENER

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<sup>3</sup> My list would include Sections 6.4, 9.1, 9.9, 12.5, 12.6, and 12.7; and Chapters 13 and 16.



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## Notation

*Absolute Continuity,  $P \ll \mu$ :* The measure  $P$  is absolutely continuous with respect to (or  $P$  is dominated by) the measure  $\mu$ . See page 7.

*Convergence in Distribution:*  $Y_n \Rightarrow Y$ . See page 131.

*Convergence in Probability:*  $Y_n \xrightarrow{P} Y$ . See page 129.

*Cumulants:*  $\kappa_{r_1, \dots, r_s}$ . See page 30.

*Derivatives:* If  $h$  is a differentiable function from some subset of  $\mathbb{R}^m$  into  $\mathbb{R}^m$ , then  $Dh(x)$  is a matrix of partial derivatives with  $[Dh(x)]_{ij} = \partial h_i(x) / \partial x_j$ .

*Floor and Ceiling:* For  $x \in \mathbb{R}$ , the floor of  $x$ , denoted  $\lfloor x \rfloor$ , is the largest integer  $y$  with  $y \leq x$ . The ceiling  $\lceil x \rceil$  of  $x$  is the smallest integer  $y \geq x$ .

*Inner Product:*  $\langle x, y \rangle$ . See page 374.

*Inverse Functions:* If  $f$  is a function on  $D$  with range  $R = f(D)$ , then  $f^{-1}$ , mapping  $2^R \rightarrow 2^D$ , is defined by  $f^{-1}(B) = \{x \in D : f(x) \in B\}$ . If  $f$  is one-to-one, the inverse function  $f^{\leftarrow}$  is defined so that  $f^{\leftarrow}(y) = x$  when  $y = f(x)$ .

*Maximum and Minimum:*  $x \wedge y \stackrel{\text{def}}{=} \min\{x, y\}$  and  $x \vee y \stackrel{\text{def}}{=} \max\{x, y\}$ .

*Norms:* For  $x \in \mathbb{R}^p$ ,  $\|x\|$  is the usual Euclidean norm. For functions,  $\|f\|_{\infty} = \sup |f|$ , and  $\|f\|_2 = [\int f^2 d\mu]^{1/2}$ . For points  $x$  in an inner product space,  $\|x\| = \langle x, x \rangle^{1/2}$ .

*Point Mass:*  $\delta_c$  is a probability measure that all of its mass to the point  $c$ , so  $\delta(\{c\}) = 1$ .

*Scales of Magnitude:*  $O(\cdot)$ ,  $O_p(\cdot)$ ,  $o(\cdot)$ , and  $o_p(\cdot)$ . See page 141.

*Set Notation:* The complement of a set  $A$  is denoted  $A^c$ . For two sets  $A$  and  $B$ ,  $AB$  or  $A \cap B$  denotes the intersection,  $A \cup B$  denotes the union, and  $A - B \stackrel{\text{def}}{=} AB^c$  will denotes the set difference. Infinite unions and intersections of sets  $A_1, A_2, \dots$  are denoted

$$\bigcup_{i=1}^{\infty} A_i \stackrel{\text{def}}{=} \{x : x \in A_i, \forall i\}$$

and

$$\bigcap_{i=1}^{\infty} A_i \stackrel{\text{def}}{=} \{x : x \in A_i, \exists i\}.$$

*Stochastic Transition Kernel:*  $Q$  is a stochastic transition kernel if  $Q_x(\cdot)$  is a probability measure for all  $x$  and  $Q_x(B)$  is a measurable function of  $x$  for every Borel set  $B$ .

*Topology:* For a set  $S$ ,  $\overline{S}$  is the closure,  $S^o$  the interior, and  $\partial S = \overline{S} - S^o$  is the boundary. See page 432.

*Transpose:* The transpose of a vector or matrix  $x$  is denoted  $x'$ .

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